COMSATS UNIVERSITY ISLAMABAD,WAH CAMPUS

CIVIL ENGINEERING

SUBJECT: ENGINEERING HYDROLOGY

CEP / ASSIGNMENT

SUBMITTED TO: ENGR SARMAD MANZOOR

GROUP MEMBERS:

FAIZAN MANSHAD (FA22-CVE-051)

KASHAN SHAFIQ (FA22-CVE-055)

FAHAD BIN RASHEED (FA22-CVE-013)

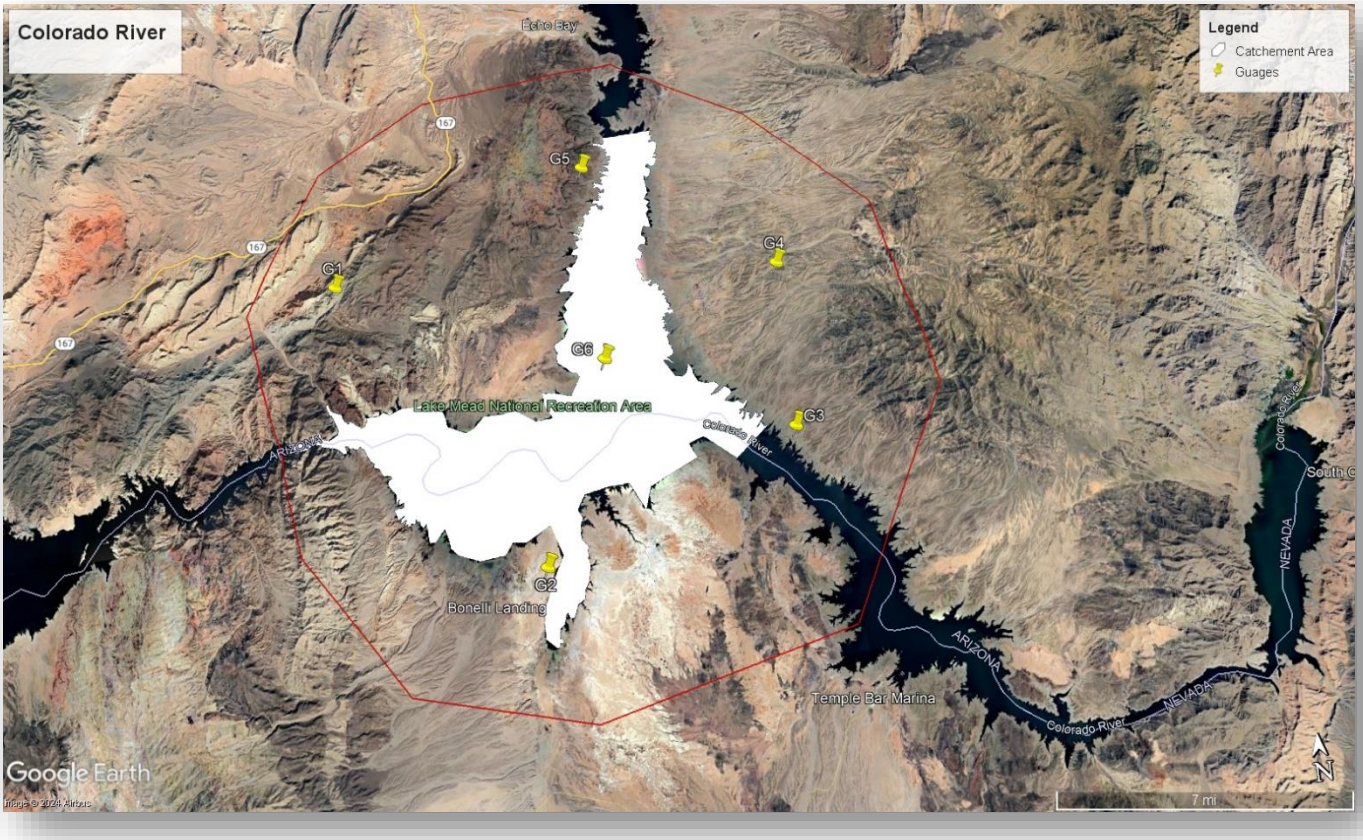
FAIZ FAREED (FA22-CVE-004)

UMER NAVEED (FA20-CVE-073)

CLASS / SECTION: 5A

DATE: /12/2024

PROBLEM STATEMENT:



Colorado River Grand Valley

A catchment area of 600 square kilometers is being studied to analyze various hydrological aspects. The region has a diverse topography and climate patterns, and the following tasks need to be carried out for the analysis:-

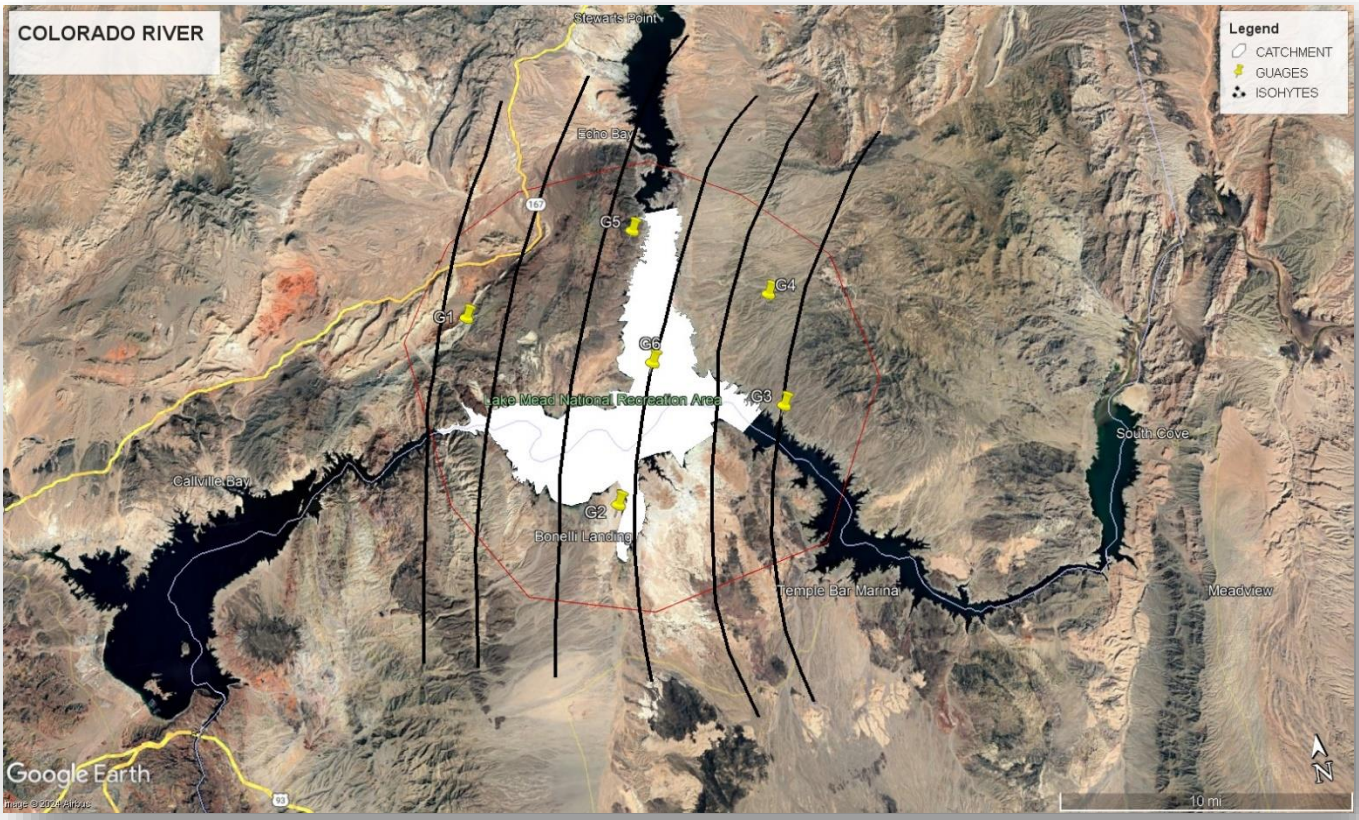
Hydrological Equation: Determine the water balance in the region using the hydrological equation ($P = R + E + \Delta S$). The annual precipitation is recorded as 1200 mm, runoff is 450 mm, and evapotranspiration is 600 mm. Calculate the change in storage (ΔS) and interpret the results.

Adequacy of Rain Gauge: There are currently 6 rain gauge stations in the catchment area. The coefficient of variation (Cv) of rainfall measurements is 20%. Determine if the existing network is adequate for an acceptable error margin of 6.5 % in the mean rainfall estimation. (Assuming the normal annual rainfall of G1, G2, G3, G4, G5 and G6 is 1100 mm, 1200 mm, 1300 mm, 1250 mm, 1320 mm and 1170 mm respectively).

Estimation of Missing Data: One of the rain gauge stations, G4, has missing data. Rainfall recorded at three neighboring stations (G1, G2, G3) for a particular month is 300 mm, 270 mm, and 330 mm, respectively. Using both the Arithmetic Mean Method and Normal Ratio

Method (assume the normal annual rainfall of G1, G2, G3, and G4 is 1100 mm, 1200 mm, 1300 mm, and 1250 mm, respectively), estimate the missing data for G4.

Averaging Precipitation over an Area: Using the provided isohyets (figure attached), calculate the average rainfall over the catchment area using Isohyetal Method.



Estimation of Runoff: The catchment consists of 80% open space, 10% impervious gravel area, and 10% dirt area. The soil type is classified as Type B, and the catchment is under wet conditions (AMC III). Using the SCS-CN method, estimate the direct runoff resulting from a 100 mm rainfall event.

Stream Gauging: The stream cross-section is divided into five vertical segments with widths of 2 m, 3 m, 4 m, and 2 m, and the respective depths are 1.5m, 2.0m, 1.8m, 1.6m, and 1.2m. Calculate the discharge using the Mean Section Method, assuming the average velocity at each depth 0.8, 1, 1.2, 0.6 and 0.9 m/s.

Unit Hydrograph: Derive the unit hydrograph for a 1 mm rainfall over the catchment lasting 2 hours with the following discharge data (hours: 0, 2, 4, 6, 8, 10, 12; discharge: 20, 50, 80, 88, 60, 30, 20). Assuming Constant Base Flow of 20.

S-Curve Hydrograph: Generate an S-curve hydrograph with a 4-hour time interval using the above unit hydrograph.

Synthetic Hydrograph: Develop a Synthetic Hydrograph using Snyder's method for a rainfall event. Use the derived 4-hour unit hydrograph, obtained from the S-curve hydrograph, to estimate W_{75} and W_{50} values. Assume necessary parameters such as lag time and peak flow coefficients as required.

Flood Routing: Perform reservoir flood routing for a flood hydrograph using the Muskingum method.